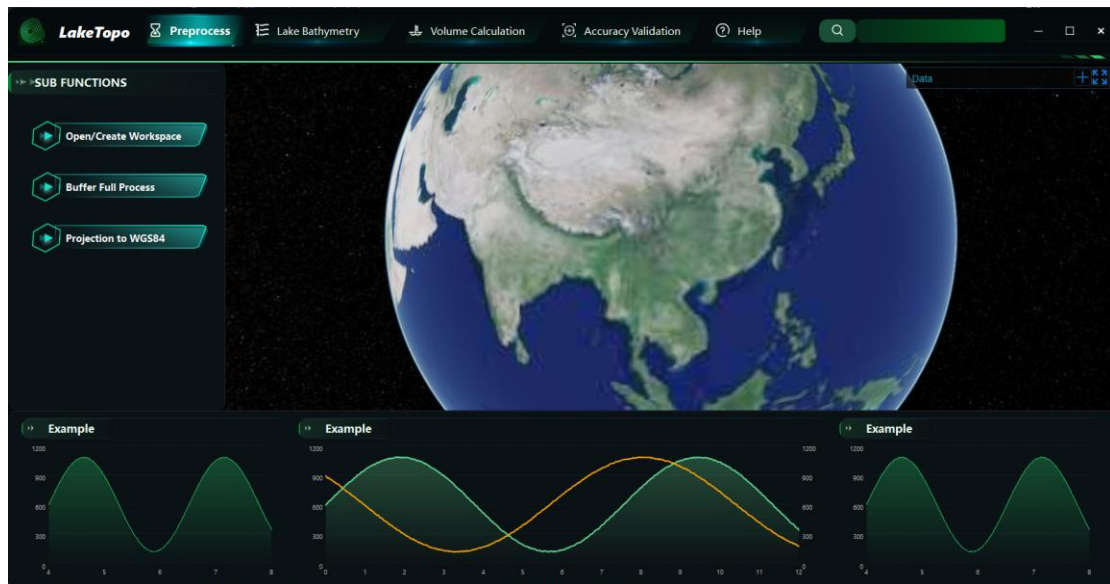


User Guide

说明:本说明文档用于 Lake Topo 软件的使用说明, 请在使用前详细阅读。由于本软件并未考虑任何商业化, 因此在软件编译上略显粗糙, 可能存在未考虑周到的逻辑问题。虽然尽量为主要的操作流程预留了空间, 但是依旧建议使用者遵循本说明内容, 避免不合理的操作引发未知错误。本软件所有生成的数据均统一至 WGS84 坐标系, 如果需要中途添加数据请确保投影统一。

This user manual is intended for the operation instructions of Lake Topo software. Please read it carefully before use. As this software was not developed with any commercial considerations, its compilation may appear somewhat rudimentary, potentially containing overlooked logical issues. While we have endeavored to accommodate the main operational procedures, users are still advised to follow the instructions herein to avoid triggering unknown errors through improper operations. All data generated by this software is uniformly projected in the WGS84 coordinate system. If you need to add data during the process, please ensure the projection is consistent. Currently, the projection conversion function has not been added, and the system is temporarily unable to automatically detect and convert projections during calculations.



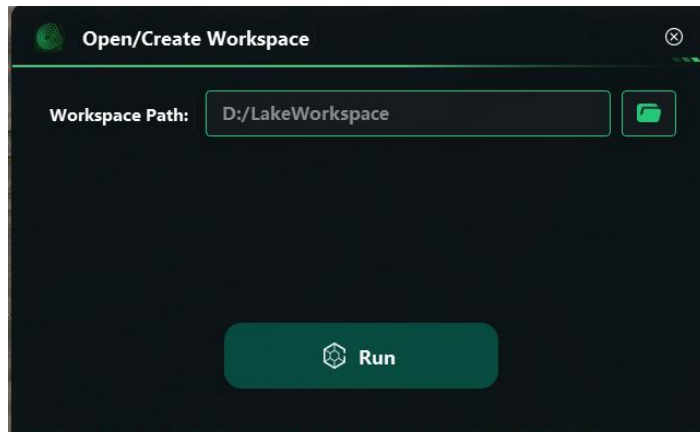
软件主界面如图所示, 目前共有 preprocess, lake bathymetry, volume calculation 和 accuracy validation 四个主模块。右上角的添加数据支持添加 tif 格式和 shp 格式的数据加载至中间的 3 维地球可视化查看, 其中 tif 格式数据需确保为正常的高程类数据以避免加载失败。搜索栏暂时未添加交互功能。接下来本指南会按照一次完整水深预测验证的流程逐模块阐明各部分使用细则

The main interface of the software is shown in the figure, currently consisting of four main modules: preprocess, lake bathymetry, volume calculation, and accuracy validation. The "

Data" button in the upper right corner supports loading data in TIF and SHP formats into the central 3D globe visualization. Note that TIF format data must be valid elevation data to avoid loading failures. The search bar currently does not have interactive functionality. This guide will subsequently explain the detailed usage of each module step by step following the complete workflow of a bathymetric prediction and validation process.

Preprocess

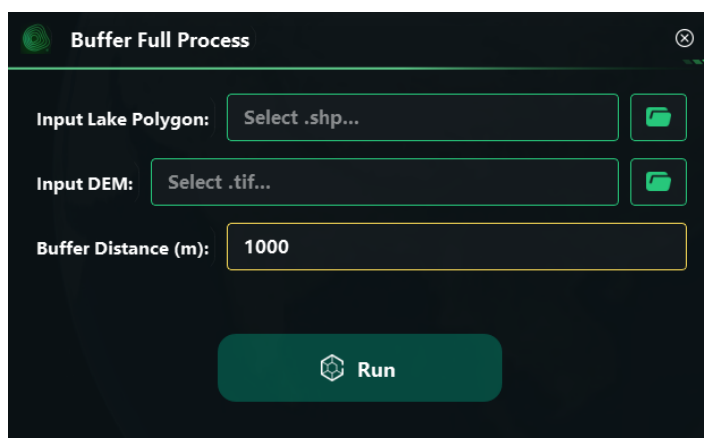
Open/create workspace



用于打开已有或创建新的工作空间，每次打开该软件时都要先确定该路径，后续所有文件处理及保存均在此路径下。需注意后续如果某些预处理没有自定义文件名输出的选项，则会将该文件夹名作为命名，例如将文件夹命名为 test1，在下一步 water extraction 中输出的水域范围 tif 会被命名为 test1_extent.tif。

Used to open an existing or create a new workspace. Each time the software is opened, this path must be determined first, and all subsequent file processing and saving will occur under this path. Please note that if some preprocessing steps do not have the option to customize the output filename, the folder name will be used as the naming convention. For example, if the folder is named "test1", the water extent TIFF output in the next water extraction step will be named "test1_extent.tif".

Buffer full process

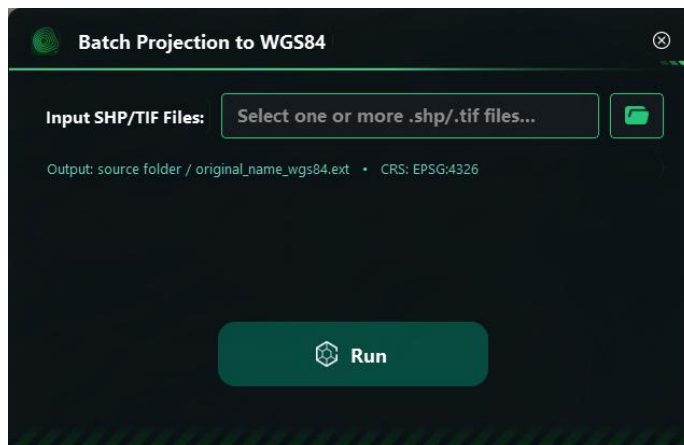


该功能输入为湖泊面 shp 和原始地表 DEM 数据，设置缓冲区宽度后输出：1.包含缓冲区的

湖泊面 shp (name_buffer.shp) 2.包含缓冲区的湖泊及周边地形 tif (name_merit.tif) 3.包含缓冲区的湖泊及周边坡度 tif (name_slope.tif) 4. 湖泊边缘线 shp (shoreline.shp) 5. 二值湖泊范围 tif (name_extent.tif)

The function takes a lake polygon SHP and the original land surface DEM data as input. After setting the buffer width, it outputs: 1. A lake polygon SHP with the buffer (name_buffer.shp); 2. A TIF file of the lake and surrounding terrain within the buffer (name_merit.tif); 3. A TIF file of the lake and surrounding slope within the buffer (name_slope.tif). 4. Lake shoreline SHP (shoreline.shp) 5. Binary lake extent TIF (name_extent.tif)

Batch Projection to WGS84

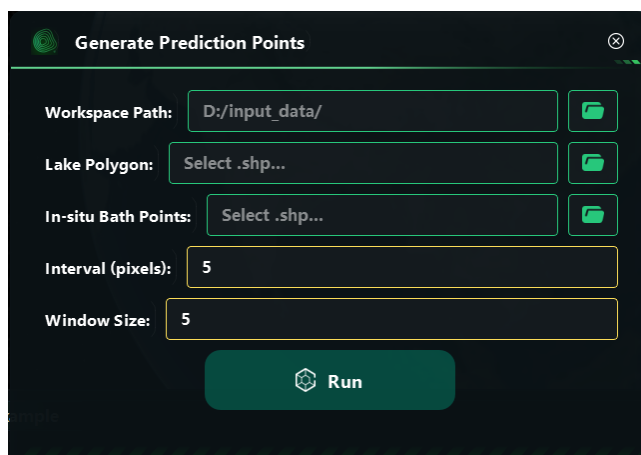


该功能帮助使用者将准备的数据中投影不一致的部分统一转换为 WGS84，通过按住 ctrl 键选择多个文件。

This feature helps users convert parts of the prepared data with inconsistent projections to WGS84 by holding down the Ctrl key to select multiple files.

Lake bathymetry

Gen. Prediction Points



该功能输入为工作空间路径，湖泊多边形 shp 以及需要事先准备的实测测深点 shp，同时前置 buffer full process 必须运行，后台会自动读取该步骤生成的 merit.tif。在设置预测点相关参数后会计算并得到：1.MLData 中用作后续训练的 txt 文件 2.tif 格式的 shoreline 和实测点文件

This function takes as input the workspace path, a lake polygon shapefile, and a pre-prepared measured bathymetric point shapefile. Additionally, the pre-processing step "buffer full process" must be run, as the system will automatically read the "merit.tif" generated in that step. After setting the parameters for prediction points, the following outputs will be computed: 1. A TXT file in MLData for subsequent training; 2. Shoreline and measured point files in TIFF format.

Gen. Prediction Depth

Generate Prediction Depth

Workspace Path:

D:/input_data/

In-situ Bath Points:

Select .shp...

Prediction Model:

XGBoost

Interval (pixels):

5

Window Size:

5

Run

该功能为机器学习生成预测点水深的主要模块，通过设置工作空间路径与实测点 shp，自动读取 MLData 中前置文件与其余相关文件，设置全湖平均分布的预测点相关参数，选择机器学习模型后在多轮优化后得到最优结果，输出 temp_ML 中的预测点 shp

This function serves as the main module for generating predicted water depths via machine learning. By setting the workspace path and the measured points SHP file, it automatically reads the prerequisite files and other related files in MLData. After configuring parameters for the prediction points evenly distributed across the entire lake and selecting a machine learning model, the optimal result is obtained through multiple rounds of optimization, and the predicted points SHP file is output to temp_ML.

Terrain generation

Water Terrain Generation

Depth Points SHP:

Select .shp...

Z Field Name:

Depth

Shoreline SHP:

Select .shp...

Lake Polygon:

Select .shp...

Output DEM:

Save as .tif...

Resolution (m):

2

Run

该功能通过利用生成的预测点 shp、湖岸线 shp、湖泊面 shp，通过自定义的分辨率实现从点到面的水深预测，输出得到水深分布的 tif 文件

This feature utilizes generated prediction point SHP files, shoreline SHP files, and lake surface SHP files to achieve depth prediction from points to surfaces with a user-defined resolution, outputting a TIF file of the water depth distribution.

Elevation adjustment



Lake Elevation Adjustment

Base Elevation (m): Optional (Leave empty)

Surrounding DEM: Select .tif...

Lake Polygon: Select .shp...

Depth Raster: Select .tif...

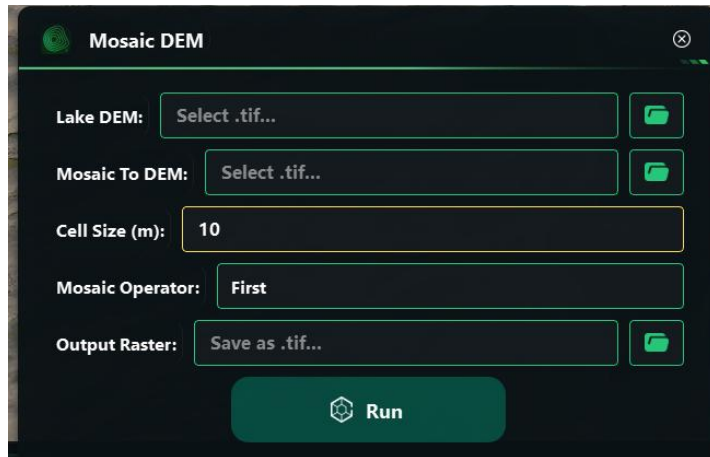
Output Raster: Save as .tif...

Run

该部分通过设置原始地形 DEM、湖泊面 shp 与上一步得到的水深 tif，通过计算得到湖盆区域的高程 DEM

This section calculates the elevation DEM of the lake basin area by setting up the original terrain DEM, the lake surface SHP, and the water depth TIF obtained in the previous step.

Mosaic DEM



Mosaic DEM

Lake DEM: Select .tif...

Mosaic To DEM: Select .tif...

Cell Size (m): 10

Mosaic Operator: First

Output Raster: Save as .tif...

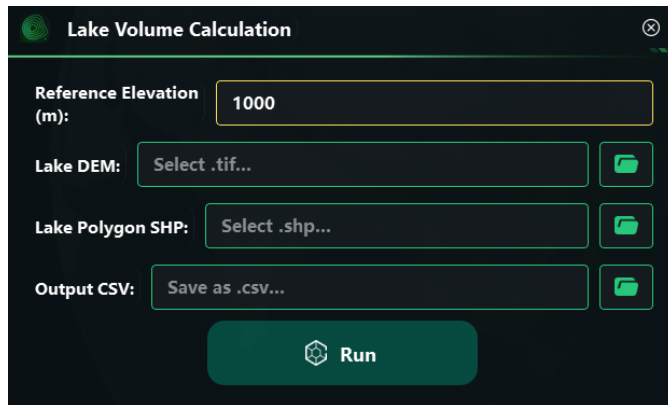
Run

该部分主要是将得到的湖盆高程 DEM 镶嵌至之前的原始地形 DEM 得到完整的湖盆及周边地区的真实地表高程 DEM，可设置分辨率与镶嵌计算方法

This section mainly involves mosaicking the obtained lake basin elevation DEM into the original terrain DEM to obtain a complete and accurate surface elevation DEM of the lake basin and its surrounding areas. The resolution and mosaic calculation method can be set.

Volume Calculation

Lake volume calculation



该部分通过输入最终的湖盆 DEM 与湖泊面 shp，计算的到该湖泊预测的水储量。如果不设置湖泊面 shp 也可设置统一的岸线高程（二者均设置的情况下会优先采用 shp）

This section calculates the predicted water storage of the lake by inputting the final lake basin DEM and the lake surface SHP. If the lake surface SHP is not set, a uniform shoreline elevation can be specified (if both are set, the SHP will take priority).

E-V Curve

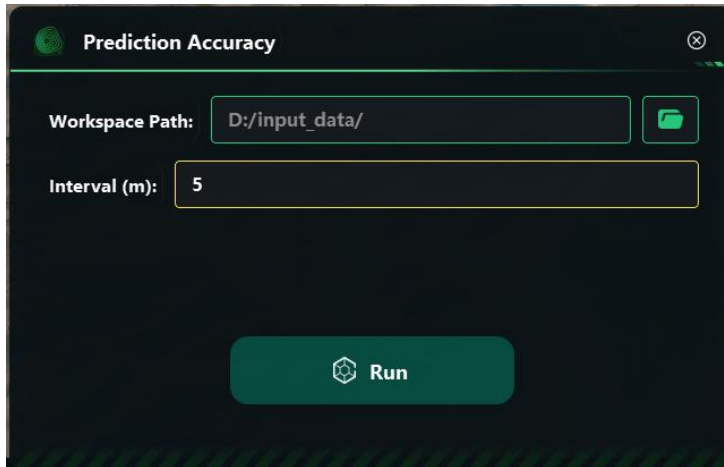


该部分通过设置湖盆 DEM 计算并绘制 E-V 图。底部可调整模拟水位来显示不同水位下的储量及 3D 模拟图

This section calculates and plots the E-V diagram by setting the lake basin DEM. The bottom allows adjusting the simulated water level to display the reserves and 3D simulation at different water levels.

Accuracy Validation

Prediction accuracy

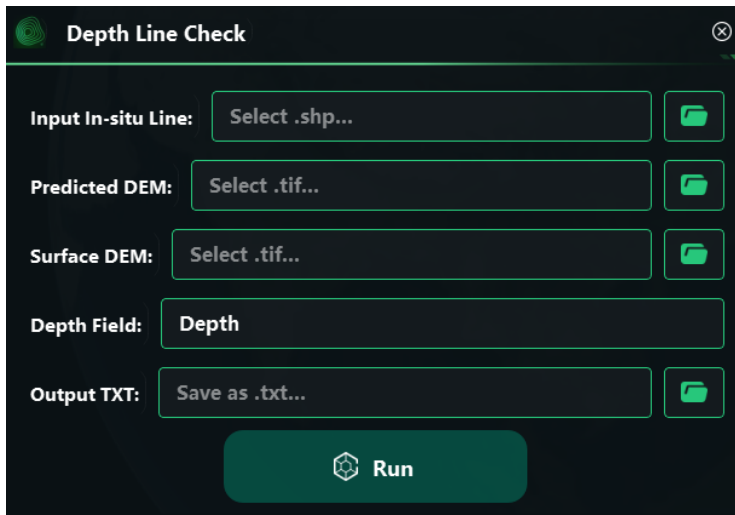


The screenshot shows a dialog box titled "Prediction Accuracy". It has a dark background with light blue text and buttons. The "Workspace Path:" field contains "D:/input_data/" and has a folder selection icon to its right. The "Interval (m):" field contains the number "5". At the bottom, there is a large blue button with a gear icon and the text "Run".

该部分通过设置工作空间路径, 运行后自动读取需要的文件(需要所有前置工作全部完成), 验证精度并输出结果 csv 与散点图

This section sets the workspace path, automatically reads the required files (all prerequisite tasks must be completed), verifies accuracy, and outputs the results in CSV format along with a scatter plot.

Depth line check



The screenshot shows a dialog box titled "Depth Line Check". It has a dark background with light blue text and buttons. There are five input fields, each with a folder selection icon to its right: "Input In-situ Line:" with "Select .shp...", "Predicted DEM:" with "Select .tif...", "Surface DEM:" with "Select .tif...", "Depth Field:" with "Depth", and "Output TXT:" with "Save as .txt...". At the bottom, there is a large blue button with a gear icon and the text "Run".

该功能通过设置实测测线 shp、预测得到的湖盆 DEM 与原始的地表 DEM, 设置实测点 shp 中水深字段名, 最后验证沿实测线的预测精度分布情况, 输出结果 txt。

This function verifies the accuracy distribution of predictions along measured lines by setting the measured survey line SHP, the predicted lake basin DEM, and the original surface DEM, and specifying the water depth field name in the measured point SHP, ultimately outputting the results in a TXT file.